Practice Set 2 , week1-3, Langman and KLM Embryo books referenced

Embryology Professional Examination: Weeks 1–3 (Conceptual Set)

1. If a mutation selectively inactivates the integrin receptors on the surface of the cell membrane of the cytotrophoblast, which initial step of development will be completely disrupted?
a. Cleavage of the zygote into blastomeres
b. Compaction of the morula cells
c. Formation of the blastocyst cavity
d. Epiblast differentiation into a bilaminar disc
e. Attachment of the blastocyst to the uterine epithelium
2. Complete primary ciliary dyskinesia caused by a genetic deficiency in dynein arms directly impairs the coordinated movement of the primitive node. This condition causes a total failure in establishing which of the following developmental events?
a. The segregation of the inner cell mass from the outer cell mass
b. Cranial closure of the neural folds
c. Gastrulation through the primitive streak
d. Left-right asymmetry of the internal organs
e. Formation of the secondary umbilical vesicle
3. In a laboratory model, primitive streak cells are chemically blocked from undergoing epithelial-to-mesenchymal transition (EMT). What structural consequence will be observed in the developing embryo?
a. Complete loss of the amnion and amniotic cavity
b. Inability of the syncytiotrophoblast to erode maternal capillaries
c. Failure of the epiblast cells to migrate and form mesoderm and endoderm
d. Failure of the bilaminar disc to form from the blastocyst
e. Complete arrest of early zygotic cleavage divisions
4. A clinical evaluation reveals that a patient's syncytiotrophoblast layers fail to develop normal lacunocanalicular networks during the second week. What is the immediate physiological consequence of this structural defect?
a. Complete failure of the zygote to undergo compaction
b. The hypoblast layer will fail to differentiate into extraembryonic endoderm
c. The amniotic cavity will expand uncontrollably into the exocoelomic space
d. The primitive streak will form prematurely at the cranial end
e. Failure to establish normal uteroplacental circulation between the mother and embryo
5. What is the fundamental embryological mechanism that explains why a sacrococcygeal teratoma often contains highly differentiated tissues such as hair, respiratory epithelium, and fully formed teeth?
a. Pluripotent primitive streak cells remain active and continue to form multiple germ layers
b. The neural crest cells undergo abnormal migration into the oral cavity structures
c. The prechordal plate forces the cloacal membrane to undergo premature rupture
d. The secondary yolk sac undergoes complete neoplastic transformation
e. Maternal decidual cells undergo abnormal metaplasia inside the uterine wall
6. If a segment of the cranial end of the notochord fails to form properly, but the caudal end develops normally, which subsequent developmental outcome will be most severely compromised?
a. The closure of the caudal neuropore

b. Formation of the vertebrae in the lumbar region
c. Induction of the forebrain and overlying neuroectoderm
d. Differentiation of the syncytiotrophoblast layer
e. Development of the lower limbs and pelvic girdle

7. During a routine ultrasound, a mass is identified obstructing the internal os of the cervix, which is diagnosed as placenta previa. What mechanical process during the first week of life caused this clinical presentation?

a. Severe scarring of the ampulla causing tubal entrapment
b. Exaggerated decidual reaction across the entire endometrium
c. Implantation of the blastocyst in the inferior segment of the uterine body
d. Premature degeneration of the zona pellucida during early cleavage divisions
e. Abnormal migration of neural crest cells into the cervix

8. If the extraembryonic somatic mesoderm fails to separate from the extraembryonic splanchnic mesoderm, which structural cavity will completely fail to develop?

a. Amniotic cavity
b. Primary yolk sac
c. Blastocyst cavity
d. Intraembryonic coelom
e. Chorionic cavity

9. In an experimental animal model, the expression of E-cadherin is completely neutralized in the blastomeres at the 8-cell stage. What morphological process will be completely arrested by this intervention?

a. Implantation into the endometrial wall
b. Compaction of the blastomeres to form a morula
c. Gastrulation through the primitive streak
d. Closure of the neural tube
e. Secretion of human chorionic gonadotropin

10. What is the primary functional significance of the decidual reaction that occurs within the endometrial stroma immediately following the attachment of the blastocyst?

a. To induce the rupture of the oropharyngeal membrane
b. To provide an immunologically privileged and nutrient-rich site that limits trophoblastic invasion
c. To stimulate the primitive streak to begin the process of gastrulation
d. To accelerate the shedding of the zona pellucida from the blastocyst
e. To induce the formation of the secondary definitive yolk sac

11. A microscopic analysis of an early embryo shows that a secondary chorionic villus has failed to transition into a tertiary chorionic villus. What specific structural component is missing from this villus?

a. A core of cytotrophoblast cells
b. Fetal blood capillaries and blood cells
c. An outer layer of syncytiotrophoblast
d. A core of extraembryonic mesoderm
e. Maternal blood sinusoids

12. If a genetic mutation prevents epiblast cells from down-regulating their cellular adhesion molecules during the third week of development, what process is directly blocked?

a. The shedding of the zona pellucida

b. Formation of the blastocyst cavity
c. Invagination of cells through the primitive streak
d. Erosion of the maternal sinusoids by the syncytiotrophoblast
e. Differentiation of the embryoblast into the bilaminar disc

13. In a case of caudal dysgenesis (sirenomelia), a patient presents with fused lower limbs and renal agenesis. What is the conceptual embryological origin of this structural defect?

a. Arrested development of the primary choriocarcinomas
b. Abnormal development of the cranial neuropore during neurulation
c. Failure of the blastocyst to lose its protective zona pellucida coat
d. Insufficient migration of mesoderm from the primitive streak to the caudal-most region
e. Failure of the hypoblast to create Heuser's membrane

14. If a chemical teratogen selectively destroys the hypoblast layer during the second week of development, which of the following structures will fail to form normally?

a. The amniotic cavity lining
b. The definitive tissues of the nervous system and skin
c. The exocoelomic membrane and the primary yolk sac lining
d. The primitive streak and its node
e. The axial skeleton and notochordal tract

15. What is the logical mechanical purpose behind the process of blastocyst "hatching" during the first week of life?

a. To allow the epiblast to expand into a trilaminar embryonic disc
b. To permit the expanding blastocyst to interact directly with the endometrial lining for implantation
c. To trigger the rhythmic pumping mechanism of the primitive heart tube
d. To allow maternal blood to enter into the trophoblastic lacunae
e. To induce the primitive streak to begin forming at the caudal end

16. If the extraembryonic coelom expands completely around the embryo but fails to form a localized focal attachment point of mesoderm, what structure will be missing?

a. The amniotic cavity
b. The secondary yolk sac
c. The primitive streak
d. The chorion laeve
e. The connecting stalk

17. What determines the exact site where the oropharyngeal membrane will form in the trilaminar embryonic disc?

a. The region where the syncytiotrophoblast first erodes a maternal sinusoid
b. The area where the epiblast and hypoblast remain tightly fused without intervening mesoderm at the cranial end
c. The site where the primitive streak reaches its maximum length before regressing
d. The location where the connecting stalk joins the extraembryonic mesoderm
e. The area where the neural crest cells first aggregate after neurulation

18. If an early embryo experiences a developmental insult that prevents the formation of the extraembryonic mesoderm during the second week, what will be the structural consequence?

a. The epiblast will fail to split into ectoderm and endoderm
b. The primitive streak will form at the exact center of the embryonic disc
c. The chorionic cavity, primary villi, and connecting stalk will fail to develop

d. The syncytiotrophoblast will turn into a hydatidiform mole
e. The blastocyst will implant exclusively in the cervical canal

19. What basic conceptual principle explains why the second week of human development is structurally designated as the "Week of Twos"?

a. The primitive streak forms two parallel columns across the disc
b. The heart tube undergoes two distinct foldings to form chambers
c. The trophoblast and embryoblast each split into two distinct structural layers
d. The embryo develops two separate neural tubes that eventually fuse
e. The somites form in two simultaneous columns along the caudal border

20. In an early pregnancy, the maternal corpus luteum is surgically removed before the 8th week without any hormonal replacement. What is the expected clinical outcome?

a. The primitive streak will shift to a cranial position
b. The embryo will continue to develop normally using uterine gland secretions
c. The placenta will automatically accelerate its production of alpha-fetoprotein
d. Spontaneous abortion occurs because the placenta cannot yet produce enough progesterone to maintain the pregnancy
e. The blastocyst will shift into an abdominal ectopic pregnancy site

21. If the primitive streak fails to regress normally during the fourth week of development and continues to proliferate, what clinical pathology is expected at birth?

a. Sirenomelia
b. Anencephaly
c. Spina bifida occulta
d. Sacrococcygeal teratoma
e. Placenta previa

22. Which germ layer is conceptually formed when migrating epiblast cells completely displace the underlying hypoblast cells during gastrulation?

a. Ectoderm
b. Mesoderm
c. Neural crest
d. Endoderm
e. Trophoblast

23. If the syncytiotrophoblast layer secretes insufficient amounts of proteolytic enzymes during the first week of development, what will be the clinical result?

a. Formation of a malignant choriocarcinoma
b. Failure of the blastocyst to embed itself within the endometrial wall
c. Premature induction of the neural folds and neural plate
d. Excessively deep implantation leading to placenta accreta
e. Accelerated expansion of the amniotic cavity

24. What is the mechanical role of the notochord during the third week of development?

a. It acts as the primary site for the exchange of maternal nutrients
b. It forms the epithelial lining of the definitive respiratory tract
c. It serves as a signaling center that induces the overlying ectoderm to differentiate into the neural plate
d. It forms the protective fluid-filled bag that cushions the embryo from mechanical trauma
e. It erodes the endometrial stroma during early implantation

25. A histopathological examination of a placenta shows chorionic villi that contain an internal mesodermal core but completely lack any visible vascular capillaries. This indicates an arrest at which stage of villous development?

a. Primary villus stage
b. Secondary villus stage
c. Tertiary villus stage
d. Stem villus stage
e. Anchoring villus stage

26. If a teratogen completely selectively blocks the formation of the primitive node (Hensen's node), which of the following structures will be absent?

a. Cytotrophoblast layer
b. Primary yolk sac lining
c. Notochordal process and axial midline structures
d. Extraembryonic somatic mesoderm
e. Amniotic membrane fluid

27. What is the physiological purpose behind the development of maternal sinusoids and lacunae inside the syncytiotrophoblast during the second week?

a. To induce the formation of the primary germ layers
b. To establish a system where maternal blood can directly nourish the avascular embryonic disc
c. To provide an exit pathway for the primitive streak cells to migrate
d. To signal the cranial neuropore to close its opening
e. To create space for the expansion of the blastocyst cavity

28. If an embryo develops a defect that prevents the migration of cells through the primitive streak, leaving the primitive streak empty, which of the following layers will completely fail to form?

a. Epiblast
b. Hypoblast
c. Intraembryonic mesoderm
d. Cytotrophoblast
e. Syncytiotrophoblast

29. What structural modification transforms a primary yolk sac into a secondary (definitive) yolk sac during the second week?

a. The complete invagination of the primitive streak cells into the cavity
b. Constriction and pinching off of a large portion of the primary vesicle by the extraembryonic coelom
c. The direct fusion of the amniotic cavity with the exocoelomic membrane
d. The invasion of maternal capillaries into the inner endodermal lining
e. The complete regression of the hypoblast layer

30. If a blastocyst lacks an inner cell mass (embryoblast) but has a completely normal outer cell mass (trophoblast), what will develop if it implants?

a. A normal embryo with missing lower limbs
b. An ectopic tubal pregnancy with anencephaly
c. A hydatidiform mole that secretes high levels of hCG
d. A sacrococcygeal teratoma at the base of the spine
e. A normal trilaminar disc with an absent amnion

Answer Key

Q 1-8	Q 9-16	Q 17-24	Q 25-30
e	b	b	d
d	b	a	b
d	a	e	e
e	c	a	c
a	b	d	b
c	c	a	a
c	e	d	e
e	e	c	c